

VBF-T1LC-CALC-01

Design Calculation Sheet

Case: t1_lc | Date: 2026-08-26

Prepared: _____ Reviewed: _____ Approved: _____

Design Calculation Sheet

Contract formula: $ED = \text{discharge energy} / \text{layer mass}$; $\text{mass} = \text{sum}(\text{thickness} \times \text{area} \times (1 - \text{porosity}) \times \text{density})$.

[input_parameters]

Parameter | Value | Unit | Source

Electrode height | 0.065 | m | Chen2020

Electrode width | 1.58 | m | Chen2020

Positive electrode thickness | 7.56e-05 | m | Chen2020

Negative electrode thickness | 8.52e-05 | m | Chen2020

Separator thickness | 1.2e-05 | m | Chen2020

Positive current collector thickness | 1.6e-05 | m | Chen2020

Negative current collector thickness | 1.2e-05 | m | Chen2020

Positive electrode porosity | 0.335 | - | Chen2020

Negative electrode porosity | 0.25 | - | Chen2020

Separator porosity | 0.47 | - | Chen2020

Positive electrode density | 3262 | kg/m³ | Chen2020

Negative electrode density | 1657 | kg/m³ | Chen2020

Separator density | 397 | kg/m³ | Chen2020

Positive CC density (Al) | 2700 | kg/m³ | Chen2020

Negative CC density (Cu) | 8960 | kg/m³ | Chen2020

Nominal cell capacity | 5.0 | Ah | Chen2020

Upper voltage cut-off | 4.2 | V | Chen2020

Lower voltage cut-off | 2.5 | V | Chen2020

Electrolyte conductivity (override) | 2.5 | S/m | formulation estimate

Cation transference number (override) | 0.6 | - | formulation estimate

Electrolyte diffusivity (override) | 4e-10 | m²/s | formulation estimate

Negative particle radius (override) | 3e-06 | m | design

Total heat transfer coefficient (override) | 60.0 | W/m²/K | design (liquid cooling)

[capacity_energy]

Quantity | Value | Unit | Source

1C discharge capacity | 5.029950574 | Ah | cell/r3_V3_1c_dfn.json

Discharge energy | 17.947732074 | Wh | cell/r3_V3_energy.json

Midpoint voltage | 3.823581498 | V | cell/r3_V3_energy.json

DC resistance | 0.003382564 | ohm | cell/r3_V3_energy.json

[energy_density]

Cell mass | 0.043454528 | kg | cell/r3_V3_energy.json

Gravimetric ED | 413.023293 | Wh/kg | cell/r3_V3_energy.json

Volumetric ED | 870.312910 | Wh/L | cell/r3_V3_energy.json

[N/P and mass]

Layer | kg/m² | g/cell | Source

Positive electrode | 0.163993788 | 16.842 | cell/r3_V3_energy.json:layer_kg_m2

Negative electrode | 0.1058823 | 10.874 | same

Positive current collector | 0.0432 | 4.437 | same

Negative current collector | 0.10752 | 11.042 | same

Separator | 0.00252492 | 0.259 | same

Total | 0.423121008 | 43.45 | sum

N/P ratio | 0.667 | - | $(33133 \cdot 0.75 \cdot 85.2 \cdot 10^{-6}) / (63104 \cdot 0.665 \cdot 75.6 \cdot 10^{-6})$